

Project Introduction

Arjun M

Project Introduction

The **Document Signature App** is a secure, full-stack web application that enables users to **upload documents, place digital signatures, share signing links, and generate legally traceable signed PDFs** — similar to platforms like **DocuSign** and **Adobe Sign**.

The app eliminates the need for physical paperwork by allowing documents to be signed electronically with full **audit trails, signer identity verification, and document integrity**. It is designed with real-world enterprise workflows in mind, including authentication, document ownership, status tracking, and signature history.

This project demonstrates how modern SaaS products handle **file security, digital trust, and collaborative workflows** at scale.

Aim of the Project

Aim of the Project

The primary aim of this project is to:

- Build a **production-ready digital signature system**
- Understand **end-to-end document lifecycle management**
- Learn how **secure SaaS platforms** are architected
- Simulate real **enterprise-grade backend and frontend workflows**
- Gain hands-on experience with **PDF processing, audit logging, and token-based access**

This project is not just a CRUD app — it models **real business logic** used in legal, HR, finance, and enterprise software.

Core Problems This App Solves

Core Problems This App Solves

- Manual document signing is **slow and error-prone**
- Physical paperwork is **hard to track and store**
- Emailing PDFs back and forth has **no auditability**
- No visibility into **who signed, when, and from where**
- Risk of document tampering after signatures

This app addresses all of these with **secure uploads, signature coordinates, immutable signed PDFs, and audit trails.**



Real-World Use Cases



Real-World Use Cases

1 Business Contracts & Agreements

- Vendors sign contracts remotely
- Status tracked as *Pending / Signed / Rejected*
- Signed documents stored securely

2 HR & Onboarding

- Offer letters
- NDA signing
- Policy acknowledgements

3 Freelancers & Agencies

- Client agreements
- Proposal approvals
- Payment authorization documents

4 Legal & Compliance Teams

- Legal documents with traceability
- Audit trails for disputes
- IP-based signer tracking

5 Education & Institutions

- Consent forms

- Admission documents
- Certificates and approvals

Key Features That Mimic Industry Standards



Key Features That Mimic Industry Standards

- JWT-based authentication
- Secure PDF upload & access control
- Drag-and-drop signature placement
- Server-side PDF modification (PDF-Lib)
- Tokenized public signature links
- Audit logs with timestamps & IP
- Status lifecycle: Pending → Signed → Rejected
- Deployment-ready architecture

These are the **exact patterns used in enterprise SaaS products**.



Industry Value of This Project



Industry Value of This Project



Why Companies Care About This Project

This project showcases **skills companies actively hire for**:



SaaS Architecture

- Multi-user authentication
- Role-based document ownership
- Tokenized external access



Security & Compliance Thinking

- JWT authentication
- Audit trail implementation
- Protected document access
- Immutable signed outputs



Backend Engineering Depth

- File handling at scale
- PDF manipulation
- Middleware logging
- Status workflows



Frontend Product Thinking

- Drag-and-drop UX
- PDF rendering

- Responsive dashboards
- Status-based UI filtering

Real Business Logic (Not Tutorials)

- Document lifecycle management
 - Signature coordinates mapping
 - Final PDF generation
 - Email/share workflows
-

What This Project Signals on Your Resume

What This Project Signals on Your Resume

- ✓ “This developer can build **enterprise-grade SaaS features**”
- ✓ “Understands **security, audit, and compliance basics**”
- ✓ “Can work with **files, PDFs, and real data flows**”
- ✓ “Not just a frontend/backend dev — but a **product-aware engineer**”

This project sits in the same category as:

- CRM systems
 - HR platforms
 - LegalTech software
 - FinTech document systems
-

Why This Is a Strong Portfolio Project

DocuSign-style apps prove you can handle real-world complexity beyond CRUD apps

Compared to todo apps or basic dashboards, this project demonstrates:

- **Stateful workflows**
- **Security-first design**
- **Cross-user interactions**
- **Document integrity guarantees**

It's the kind of project that **stands out immediately** in interviews.



2-Week Build Plan



2-Week Build Plan



Week 1: Core Features, Backend & Frontend Setup

Day 1: Project Setup & Repo Initialization

- Create GitHub repo with MERN folder structure
- Initialize React app with Tailwind CSS
- Setup Node.js + Express + MongoDB (Mongoose)
- Install necessary libraries (Multer, PDF-Lib, bcrypt, JWT)
Skills: Project scaffolding, Tailwind setup, server setup

Day 2: Auth System (JWT)

- User model: name, email, password
- Routes: /register, /login
- JWT auth + bcrypt password hashing
- Auth middleware for protected routes
Skills: JWT, secure user auth, API routes

Day 3: File Upload API

- Setup Multer for PDF uploads
- Store file path & metadata in MongoDB
- Route: /api/docs/upload
Skills: File handling, middleware, API design

Day 4: View & List Documents

- API to fetch user's uploaded files
- Display files in a dashboard
- Add preview functionality using `react-pdf`
Skills: RESTful APIs, PDF rendering in React

Day 5: Signature Schema & Logic

- Signature Model: field, coordinates, signer, status
- Route to save signature positions (x, y)
- Display signature placeholder on PDF
Skills: Schema relations, position-based rendering

Day 6: PDF Editor Integration

- Add drag-and-drop signature field (HTML overlay)
- Save coordinates relative to page
- Route: POST /api/signatures
Skills: Coordinate systems, frontend UX logic

Day 7: Buffer / Testing

- Debug UI and backend integration
 - Write Postman tests for API
-

✓ Week 2: Signature Rendering, Sharing & Polish

Day 8: Generate Final Signed PDF

- Use **PDF-Lib** to embed signature text/image
- Export signed version to disk/cloud
Skills: PDF processing, server-side document generation

Day 9: Email + Public Signature Links

- Generate tokenized URL for external signature
- Email link to signer (use nodemailer or mock)
Skills: Token auth, email logic

Day 10: Audit Trail

- Log who signed, when, and IP (use middleware)
- Route: GET /api/audit/:fileId
Skills: Middleware logging, basic audit management

Day 11: Signature Status Updates

- Status: Pending, Signed, Rejected
- Allow signer to accept/reject with reason
Skills: Status flows, conditional logic

Day 12: Dashboard UI Polish

- Filter by signature status
- Responsive layout with Tailwind
Skills: Tailwind UI polish, component reuse

Day 13: Deployment

- Backend: Render / Railway
- Frontend: Vercel / Netlify
- MongoDB Atlas for DB

Day 14: Final Testing + Demo

- Prepare a GitHub README
- Record a 2-min demo walkthrough

Web Tech Stack for Document Signature App

Web Tech Stack for Document Signature App

Recommended Tech Stack (Modern & Industry-Ready)

Frontend

- **React (Vite or Next.js)**
- **TypeScript** (highly recommended)
- **Tailwind CSS**
- **react-pdf**
- **dnd-kit** (drag & drop)
- **Axios**
- **React Hook Form**
- **Zod** (validation)

Backend (API Layer)

- **Node.js + Express**

Auth

- **JWT (access + refresh tokens)**
- **Protected routes**

Storage

- **Local storage (dev)**
- **Supabase Storage / AWS S3 (prod)**

Deployment

- **Frontend:** Vercel / Netlify
- **Backend:** Render / Railway
- **DB:** Supabase / MongoDB Atlas

API Endpoints & Mock Data

API Endpoints & Mock Data

Auth APIs

POST /api/auth/register

POST /api/auth/login

Document APIs

POST /api/docs/upload // Upload PDF

GET /api/docs/ // List user PDFs

GET /api/docs/:id // View specific doc

Signature APIs

POST /api/signatures // Save signature position

GET /api/signatures/:id // Get signatures for document

POST /api/signatures/finalize // Embed signature into PDF

Audit APIs

GET /api/audit/:docId

Mock Data: User

```
{
  "name": "Alice",
  "email": "alice@example.com",
  "password": "hashedpassword"
}
```

Mock Data: Signature

```
{
  "documentId": "abc123",
  "userId": "xyz456",
  "x": 120,
  "y": 330,
  "page": 1,
  "status": "signed"
}
```



Skills You Will Learn by Stage



Skills You Will Learn by Stage

Week	Skill	Description
1	Auth & API	JWT, bcrypt, REST APIs
1	File Upload	Multer, server-side storage
1-2	Frontend UX	Drag-and-drop UI, react-pdf
2	PDF Generation	pdf-lib for embedding data
2	Security	Tokenized links, protected routes
2	Audit Logging	Middleware, action tracking
2	Deployment	Hosting full-stack app

Resources

Learning Resources

1. PDF-Lib JS Docs: <https://pdf-lib.js.org/docs>
 2. MERN Auth (JWT): <https://www.youtube.com/watch?v=2jqok-Wgell> (PedroTech)
 3. Multer File Upload: <https://www.youtube.com/watch?v=9Qzmri1WaaE>
 4. PDF Signature with Node:  Signing a PDF with Javascript - Going through the code
 5. React PDF Viewer:  How to View PDFs in React JS with React PDF Viewer
 6. React Drag and Drop Signatures:
 React Drag And Drop (dnd-kit) | Beginners Tutorial
-

Web Tech Stack Resources (Step-by-Step)

1. React (Core)

 YouTube

React Full Course (Beginner → Advanced)

 <https://www.youtube.com/watch?v=bMknfKXIFA8>

 **Covers:**

- Hooks
- State
- Components
- API calls

 Docs

- React Official Docs
👉 <https://react.dev/>

⚡ 2. Vite / Next.js Setup

 YouTube

Vite + React Setup

👉 https://www.youtube.com/watch?v=GgMbL0z_h8A

Next.js App Router (Optional)

👉 <https://www.youtube.com/watch?v=wm5gMKuwSYk>

🎨 3. Tailwind CSS (UI Polish)

 YouTube

Tailwind CSS Crash Course

👉 <https://www.youtube.com/watch?v=dFgzHOX84xQ>

✅ Perfect for dashboards

Docs

- Tailwind Docs

👉 <https://tailwindcss.com/docs>

4. PDF Rendering (CRITICAL)

YouTube

View PDF in React using react-pdf

👉 <https://www.youtube.com/watch?v=Uu4W8r9sKxQ>

Docs

- react-pdf Docs

👉 <https://github.com/wojtekmaj/react-pdf>

5. Drag-and-Drop Signatures (CORE FEATURE)

YouTube

Drag & Drop using dnd-kit

👉 <https://www.youtube.com/watch?v=FZt2N6R0YkY>

Docs

- dnd-kit Docs

👉 <https://docs.dndkit.com/>

6. API Integration (Axios + JWT)

 YouTube

Axios Crash Course

👉 <https://www.youtube.com/watch?v=6LyagkoRWYA>

JWT Auth in React

👉 <https://www.youtube.com/watch?v=nDGA3km5He4>

 Docs

- Axios Docs

👉 <https://axios-http.com/>

7. Auth & Protected Routes

 YouTube

Protected Routes in React

👉 <https://www.youtube.com/watch?v=2k8NleFjG7I>

8. Forms & Validation

 YouTube

React Hook Form + Zod

👉 <https://www.youtube.com/watch?v=Z1Nw5FfKqC8>

 Docs

- React Hook Form
👉 <https://react-hook-form.com/>
 - Zod
👉 <https://zod.dev/>
-

9. Coordinate System (IMPORTANT)

Concept to Learn

- Convert absolute → relative coordinates
- Save (x %, y %)
- Re-render correctly on different screens

 Video

Mouse Position & Bounding Box in React

👉 <https://www.youtube.com/watch?v=qd5y8T7Zx9U>

10. Deployment

 YouTube

Deploy React App on Vercel

👉 <https://www.youtube.com/watch?v=2h6n5Dgk0oI>

🏆 Recommended Learning Order

St ep	Learn
----------	-------

1	React basics
---	-----------------

2	Tailwind
---	----------

3	react-pdf
---	-----------

4	Drag & Drop
---	----------------

5	Axios + JWT
---	----------------

6	Protected routes
---	---------------------

**7 Coordinat
 e logic**

8 UI polish

**9 Deploym
 ent**

 Why This Web Stack Is Perfect

This tech stack matches real SaaS dashboards used in industry

You'll be learning:

- **Component-driven UI**
- **Drag-and-drop UX**
- **Secure frontend auth**
- **PDF rendering workflows**
- **Responsive enterprise UI**

